

Wen Rui Sun

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University of Alberta
Department of Mathematical and Statistical Sciences

Research Interests

Convex Geometry, Combinatorial Geometry, Geometric Tomography, Functional Analysis

Education

University of Alberta, Edmonton, Canada

PhD in Pure Mathematics	2022–Expected Aug. 2027
Supervisors: Vladyslav Yaskin and Beatrice-Helen Vritsiou	
MSc in Pure Mathematics	2020–2022
Thesis: Illumination of Convex Bodies with Symmetries in Dimensions 3 and 4	
Supervisors: Vladyslav Yaskin and Beatrice-Helen Vritsiou	
BSc in Mathematics (Specialization)	2018–2020
BSc in Petroleum Engineering	2013–2018

Publications

3. **On Axial Symmetry in Convex Bodies**
with *R. Goenka*, *K. Moore*, and *E. P. White*. *Pacific J. Math.* **341** (2026), no. 2, 275–303.
[doi:10.2140/pjm.2026.341.275](https://doi.org/10.2140/pjm.2026.341.275)
2. **On the illumination of 1-symmetric convex bodies**
with *B.-H. Vritsiou*. *Pure Appl. Funct. Anal.* **10** (2025), no. 6, 1483–1524.
[arXiv:2407.10314](https://arxiv.org/abs/2407.10314)
1. **Illuminating 1-unconditional convex bodies in $\mathbb{R}^3$ and $\mathbb{R}^4$, and certain cases in higher dimensions**
with *B.-H. Vritsiou*. *Canad. J. Math.* (2025), 1–67.
[doi:10.4153/S0008414X25101260](https://doi.org/10.4153/S0008414X25101260)

Preprints

1. **On determination of convex bodies by their section functions**
with *A. Fish*, *D. Ryabogin*, and *V. Yaskin*. Preprint, 2026.
[arXiv:2610.10229](https://arxiv.org/abs/2610.10229).

Talks

6. **Illumination of Symmetric Convex Bodies and Some Non-Symmetric Generalizations**
at *AMS Spring Central Sectional*, BIRS, Banff, Alberta, Canada. 2026.
5. **Illumination of 1-Symmetric Convex Bodies**
at *AMS Spring Central Sectional*, NDSU, Fargo, North Dakota, USA. 2026.
4. **Applications of Deep Illumination for the Illumination of 1-Unconditional Bodies**
at *AMS Fall Central Sectional*, Saint Louis University, Saint Louis, Missouri, USA. 2025. (Delivered online).
3. **Measure of Axiality for Convex Figures**
at *CMS Winter Meeting*, Montreal, Quebec, Canada. 2023.
2. **Illumination Conjecture for Convex Bodies with many Symmetries**
at *Harmonic Analysis and Convexity*, BIRS, Banff, Alberta, Canada. 2023.
1. **Illumination Conjecture for Convex Bodies with many Symmetries**
at *Alberta Mathematics Dialogue*, Mount Royal University, Calgary, Alberta, Canada. 2023.

Poster Presentations

1. Illumination of 1-Symmetric Convex Bodies

at *Recent Advances in Applications of Harmonic Analysis to Convex Geometry*, Kent University, Kent, Ohio, USA. 2023.

Awards

Alberta Graduate Excellence Scholarship, University of Alberta	2025–2026
Alberta Graduate Excellence Scholarship, University of Alberta	2024–2025
First Class Standing, University of Alberta	2018–2019
First Class Standing, University of Alberta	2013–2014
University of Alberta Academic Excellence Scholarship	2013
Alexander Rutherford Scholarship	2010–2013
International Baccalaureate Certificate	2013
Outstanding Honor Student, Sir Winston Churchill High School	2013

Teaching Experience

Graduate Teaching Assistantships, University of Alberta, Edmonton, AB Sept 2020–Curr.

- **Teaching Courses**
MATH 154 - Calculus for Business and Economics I.
- **Teaching Labs**
Responsible for teaching course material, administering lab quizzes, and assisting students with homework and conceptual questions.
- **Decima Robinson Centre drop-in help**
Provide one-on-one academic support to students seeking help with course material, explaining concepts and guiding problem-solving across various subjects.
- **Grading**
- **Exam Proctoring**